

Cache Controller Architecture Description

Project Overview

This document explains the architecture of a custom RTL direct-mapped cache controller implemented in Verilog. The cache subsystem interfaces between a CPU and external main memory.

Major Blocks

The design consists of Address Decoder, Cache Controller, Cache Datapath, and Cache Memory modules. The top module integrates these blocks.

Address Decoder

The CPU address is divided into tag, index, word offset, and byte offset fields. The decoder provides the tag, index, and word offset required by the cache (byte offset is not used in this project for convenience).

Cache Memory

The cache array stores data words along with tag, valid bit, and dirty bit metadata. The valid bit indicates line availability and the dirty bit indicates modification compared to memory.

Controller and Datapath Separation

The controller decides the operation using an FSM. The datapath performs address generation, data routing, and multiplexing according to controller signals.

Interfaces

The CPU interface supports read/write transactions. The memory interface supports refill and dirty line writeback using a custom handshake protocol.